L-P-527B
August 8, 1973
SUPERSEDING
Fed. Spec. L-P-527A
June 24, 1966

FEDERAL SPECIFICATION

PLASTIC SHEET, STYRENE-BUTADIENE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for sheets made from molding materials conforming to the requirements of L-P-398.

1.2 Classification.

1.2.1 Types, Grades and Classes. Plastic sheets conforming to this specification shall be of the following types, grades and classes, as specified (see 6.2):

Type I - Lowest impact strength.
Type II - Moderate impact strength.
Type III - Higher impact strength.
Type IV - Highest impact strength.
Grade A - Lower Vicat softening point.
Grade B - Higher Vicat softening point.
Class 1 - Low orientation.
Class 2 - Medium orientation.
Class 3 - High orientation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 9330

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Federal Specifications:

L-P-398 - Plastic Molding Material, Styrene Butadiene.
UU-P-268 - Paper, Kraft, Untreated, Wrapping.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies).

Military Specification:

MIL-L-10547 - Liners, Case and Sheet Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing and Materials (ASTM) Standards:

D 618-61 (1971) - Conditioning Plastics and Electrical Insulating Materials for Testing.

D 883-72c - Plastics, Def. of Terms Relating to Standard Nomenclature Relating to Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W. Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Sheets shall be manufactured from styrene-butadiene molding material conforming to L-P-398. Sheets shall be made by extrusion molding and press polished (see 3.9.2), and shall be annealed if this operation is necessary to assure compliance with the orientation requirement of 3.3. The sheet manufacturer shall state which material (type, grade, trade name and manufacturer's number, including color number) was used to make each lot of sheets.

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3.2 Form. The material shall be furnished in the form of flat rectangular sheets of the dimensions specified.

3.3 Orientation. The sheets, when tested as specified in 4.2.3, shall conform to the requirements of table I.

TABLE I. Orientation (percent shrinkage)

Sheet thickness	Class 1	Class 2	Class 3
Inch	Percent	Percent	Percent
0.010 - 0.019	Greater than 30	30 - 60	Greater than 60
.020 - .039	Greater than 20	20 - 50	Greater than 50
.040 - .059	Greater than 20	20 - 40	Greater than 40
.060 - .079	Greater than 10	10 - 30	Greater than 30
.080 - 1.00 and over	Greater than 5	5 - 20	Greater than 20

3.4 Impact resistance. When considered necessary by the procuring agency, minimum acceptable impact resistance and a corresponding method of test shall be specified by the procuring agency (see 6.2).

3.5 Warp or twist. Applicable only to sheets having diagonals 36 inches or more in length. The maximum deviation from a 36 inch straight edge shall not exceed the values shown in table II, when measured in accordance with 4.3.2.

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TABLE II. Warp or twist

Thickness of sheet, inches	Maximum deviation from 36 inch straight edge, Inch
Up to 1/8	0.40
Over 1/8 to 1/4, incl	.250
Over 1/4 to 3/4	.18
Over 3/4	.10

3.6 Color. The color shall be as specified by the procuring agency (see 6.2).

3.7 Dimensions and tolerances.

3.7.1 Width and length. Unless otherwise specified (see 6.2), the standard size of sheets shall be 20 inches by 40 inches and the tolerance on each dimension shall be:

Width - plus or minus 1/8 inch
Length - plus 1/4, minus 0 inch

3.7.2 Thickness. The thickness shall be as specified in the contract or order within a tolerance of plus 10 percent, minus 5 percent of the specified thickness. The minimum allowable thickness shall be 0.010 inch.

3.8 Instruction sheet. Each lot of sheets shall be accompanied by instruction sheets (the number to be specified by the procuring agency) giving directions for forming. These directions shall include forming temperature, the maximum amount of draw, and the minimum bending radius to which the sheets should be subjected, together with any additional information specified in the order, or considered desirable by the supplier.

3.9 Workmanship.

3.9.1 Uniformity. All sheets in the lot shall, when subjected to visual examination, satisfy the color and surface finish requirements of this specification.

3.9.2 Surface finish. Unless otherwise specified, the surface shall be press polished free from defects such as pits, blisters, cracks, dents, waviness, heat marks, and scratches. (These defects shall be defined in accordance with ASTM D 883-72c as applicable.)

3.9.3 Internal defects. Edges and cut sections shall show no porosity or delaminations.

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3.10 Cover sheets for packaging. Unless otherwise specified by the procuring agency, all material shall be separated by size and thickness when packaged for shipment. Individual sheets shall be covered on both sides by a suitable adhered paper or film that can be readily removed without injury to the surface, and that will adequately protect the surface from scratching or damage during handling, shipping or storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, grade, class, and thickness submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified. In the event of conflict, this specification shall govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of sheets.

4.2.2.1 Examination of sheets. Examination of sheets shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of plastic sheets for examination in 4.2.2.1.1, 4.2.2.1.3, 4.2.2.1.4, and in units of shipping containers for examination in 4.2.2.1.5.

4.2.2.1.1 Examination of sheets for color. The sample unit for examination of color of sheets shall be one plastic sheet. The color shall conform to the requirements of 3.6 when tested as specified by the procuring agency.

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4.2.2.1.2 Examination for instruction sheets. The sample unit for examination for instruction sheets shall be one lot of sheets. The number of instruction sheets shall be as specified by the procuring agency (see 3.8 and 6.2). All instruction sheets shall conform to the requirements of 3.8.

4.2.2.1.3 Examination of sheets for defects in dimensions. The sample unit for the examination specified in table III shall be one plastic sheet.

TABLE III. Examination of sheets for defects in dimensions

Examine	Defects
Width	Varies by more than plus or minus 1/8 inch from width specified.
Length	Varies by more than plus 1/4, minus 0 inch from length specified.
Thickness	Varies by more than plus 10 percent or minus 5 percent from the specified thickness.

4.2.2.1.4 Examination of sheets for defects in appearance and workmanship. The sample unit for the examination specified in table IV shall be one plastic sheet.

TABLE IV. Examination of sheets for defects in appearance and workmanship

Examine	Defect
Appearance	Not uniform texture, finish or color. Any pits, blisters, cracks, dents, waviness, heat marks, or scratches.
Workmanship	Surface not press polished and straight; edges not smooth and straight. Any delaminations or porosity on edges.

4.2.2.1.5 Examination of the preparation for delivery requirements. An examination shall be made in accordance with Table V to determine that packaging, packing and marking comply with requirements specified in section 5. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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TABLE V. Examination for defects in preparation for delivery

Examine	Defects
Cover sheets for packaging	Individual sheets not completely covered on both sides with adhered paper or film, as specified (see 3.10).
Packaging	Not packaged as specified. Not level specified.
Packing	Any nonconforming component, component missing, damaged, or otherwise defective affecting serviceability. Not level specified. Materials and construction not as specified.
Count	Less than specified or indicated quantity.
Marking	Interior or exterior markings omitted, illegible, incorrect, imcomplete, of improper size, location, sequence, method of application, or not in accordance with contract requirements.

4.2.2.1.6 Inspection levels and AQLs for examinations. The inspection levels for determining the sample size and the AQL, expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	I	1.5
4.2.2.1.2	S-3	2.5
4.2.2.1.3	S-2	2.5

4.2.3 Testing. The sheet shall be tested for the applicable characteristics listed in tables I and II in accordance with the test methods specified herein, for each lot submitted for inspection. The lot size, for the purpose of determining sample size for testing, shall be expressed in units of 50 pounds of sheets. The sample unit shall consist of sufficient material to prepare all specimens required for testing. The inspection level shall be S-1 with an acceptance number of 0. The results of each test shall be the averaged results of the specimens.

4.3 Test Methods.

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4.3.1 Specimens. Test specimens shall be prepared from the sheets sampled in accordance with 4.2. No more than one specimen for each test, except impact resistance, shall be prepared from any sheet sampled, except where the sample size is less than five sheets. In such cases, specimens shall be selected from each sheet for each test. Unless otherwise specified, testing shall be 23° ± 1°C (73.4° ± 1.8°F.) and 50 ± 5 percent relative humidity, after conditioning in accordance with procedure A of ASTM D 618-61(1971).

4.3.2 Orientation. Orientation shall be determined on five specimens, each 4 inches square. A small circle with an arrow inclosed shall be inscribed in one corner. The arrow shall indicate the direction of extrusion. An oil bath filled with Humble Primol 355 or equivalent shall be stabilized at a temperature of 152° ± 3°C. (305° ± 5°F.). Specimems under 0.100 inch thickness shall be immersed in oil bath for a period of 10 minutes, plus 1/2, minus 0 minutes. Specimens over 0.100 inch in thickness shall be immersed for a period of 16 minutes, plus 1/2 minus 0 minutes. The specimen shall then be removed, cooled, and remeasured. Calculate percent shrinkage as follows:

$$\text{Shrinkage, percent} = \frac{\text{Original length} - \text{Final length}}{\text{Original length}} \times 100$$

4.3.3 Warp or twist. Two sheets shall be tested in accordance with method 6051 of Fed. Test Method Std. No. 406 except for the following: A straight edge 36 inches long shall be used. The straight edge shall be moved over the sheet and rotated along the sheet length about an axis perpendicular to the plane of the sheet without permitting an end of the straignt edge to extend beyond the edge of the sheet. The maximum distance, in inches, from the sheet to the straight edge shall be measured and reported. The measurement shall be made with the sheet lying with its surface of maximum convexity in contact with a horizontal flat surface. The straight edge shall be light in weight and no pressure shall be exerted on it which might tend to reduce the warp or twist during the measurement.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Sheet material shall be interleaved in accordance with 3.10 and shall be overwrapped with kraft wrapping paper conforming to UU-P-268 and tightly sealed with tape conforming to type III, grade B of PPP-T-45.

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5.1.2 Level C. Sheets shall be packaged to afford protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Packages of sheets shall be packed in shipping containers conforming to PPP-B-585 (class 3), PPP-B-601 (overseas type), or PPP-B-621 (class 2). Unless otherwise specified, containers shall be provided with a case liner conforming to MIL-L-10547. Boxes shall be closed, strapped, or banded in accordance with the applicable box specifications or appendix thereto. The gross weight of wirebound wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.2 Level B. Packages of sheets shall be packed in shipping containers conforming to PPP-B-585 (class 1), PPP-B-601 (domestic type), or PPP-B-621 (class 1). Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. The gross weight of wirebound wood or wood-cleated containers shall not exceed 200 pounds.

5.2.3 Level C. Packages of sheets shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

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6.1 Intended use. Sheets conforming to this specification are intended for use in instrument housings and applications requiring resistance to impact. Methods of fabrication may require that the sheet be drawn or bent. Type I is intended for general purpose applications requiring toughness and chemical resistance. Type II is intended for applications which require somewhat greater shock resistance and durability than type I. Types III and IV are intended for usage which requires greater elongation and impact strength than is obtainable with either type I or II material. Grades A and B are specified in accordance with deflection temperature under load behavior. Class 1, 2, and 3 are specified in accordance with shrinkage requirements (see 6.3).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification

(b) Type, grade and class of material required (see 1.2, and L-P-398).

(c) When required, impact resistance values and method of test (see 3.4)

(d) Length and width of sheets (see 3.7.1).

(e) Thickness of sheet (see 3.7.2)

(f) Color and method of test (see 3.6)

(g) Number of instruction sheets required (see 3.8)

(h) Selection of the applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2)

(i) Quantities of plastic sheets required in the unit packages (see 5.1.1 and 5.1.2)

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6.3 Sheet shrinkage. Shrinkage in the sheets is an indication of orientation. The higher the orientation, the higher the shrinkage.

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Review activities:
Army - GL, EL, MI
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CIVIL AGENCY COORDINATING ACTIVITY:
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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION L-P-527B, Plastic Sheet, Styrene-Butadiene

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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